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Credential, and Selectivity

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Source: *Demography*, Vol. 36, No. 4 (Nov., 1999), pp. 445-460

Published by: Duke University Press on behalf of the Population Association of America

Stable URL: <https://www.jstor.org/stable/2648083>

Accessed: 28-04-2026 17:20 UTC

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REFINING THE ASSOCIATION BETWEEN EDUCATION AND HEALTH: THE EFFECTS OF QUANTITY, CREDENTIAL, AND SELECTIVITY*

CATHERINE E. ROSS AND JOHN MIROWSKY

We refine the established association between education and health by distinguishing three aspects of a person's education (quantity, credential, and selectivity) and by examining the mechanisms through which they may correlate with health. Data are from the 1995 Aging, Status, and the Sense of Control Survey, a representative U.S. national telephone survey of 2,593 respondents aged 18 to 95, with an oversample of elderly. Results show that physical functioning and perceived health increase significantly with years of formal education and with college selectivity for those with a bachelor's or higher degree, adjusting for age, sex, race, marital status, and parental education. The credential of a college degree has no net association with physical functioning and perceived health beyond the amount attributable to the additional years of schooling. Of the three aspects of education, years of schooling has the largest effect. Most of that association appears attributable to its correlation with work and economic conditions, social psychological resources, and health lifestyle. A large portion of the net association of college selectivity with physical functioning and perceived health appears attributable to health lifestyle.

Educational attainment has positive effects on health. The well educated experience better health than do the poorly educated, as indicated by high levels of perceived health and physical functioning and low levels of morbidity, mortality, and disability (Doornbos and Kromhout 1990; Feldman et al. 1989; Fox, Goldblatt, and Jones 1985; Guralnik et al. 1993; Gutzwiller et al. 1989; Kaplan, Haan, and Syme 1987; Leigh 1983; Liu et al. 1982; Morris 1990; Pappas et al. 1993; Ross and Wu 1995; Syme and Berkman 1986; Williams 1990; Winkleby et al. 1992; Woodward et al. 1992).

We refine this association by asking two questions about health. First, what aspect of education affects health: quantity, selectivity, or credentials? Most research measures only the amount of education received. Yet two people with the same amount of education may have different experiences and outcomes. We measure quantity of education by years of formal education completed; credentials by whether one has a college degree; and, for the college educated, selectivity

by the quality or prestige of one's college or university. Each aspect of education may or may not correlate with health net of the others. Second, what are the mechanisms by which education (in terms of quantity, selectivity, or credential) affects adult health? We focus on three paths by which education may affect health: work and economic conditions, social psychological resources of perceived control and social support, and health lifestyle.

To date, no research has separated and compared these three aspects of schooling. Thus, we first develop three models of the processes by which each aspect could affect health: the quantity model, the credential model, and the selectivity model. We examine two aspects of health: physical functioning and perceived health.

THREE MODELS OF EDUCATION AND HEALTH

The Quantity Model

According to human capital theory and status attainment models, schooling teaches people skills that are useful for work and life in industrial and postindustrial economies (Becker 1964; Hyman, Wright, and Reed 1976; Sewell and Hauser 1976; Spaeth 1976). The more years of schooling, the more human capital acquired (Becker 1964). Schooling provides general skills, especially cognitive skills, and specific skills useful for work; it socializes people in the values, behaviors, and dispositions necessary for achievement. The more years of schooling, the greater the cognitive development, characterized by flexible, rational, complex strategies of thinking (Hyman et al. 1976; Nunn, Crockett, and Williams 1978; Spaeth 1976). Higher education teaches people to think logically and rationally, to see many sides of an issue, and to analyze and solve problems (Kohn and Schooler 1982; Kohn and Slomczynski 1993). Furthermore, higher education teaches people the cognitive skills necessary to continue learning out of school (Hyman et al. 1976; Hyman and Wright 1979). Schooling also increases specific skills and knowledge, like biochemistry, math, journalism, history, and statistics, which may be less important than general cognitive skills but still may be relevant. Finally, schooling shapes dispositions, behaviors, and personality traits, like self-directedness and an orientation toward hard work. In school, people are socialized to be dependable, to use good judgment, to be self-motivated, to have self-confidence, to value hard work, to trust coworkers, to plan, and so on (Kohn and Slomczynski 1993). The more years of schooling, the greater a person's stock of human capital.

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Spaeth (1976) depicts the process of status attainment as one in which years of schooling captures an individual's exposure to increasingly complex environments, which leads to increased cognitive skills. A number of empirical studies support the exposure model. Years of schooling has enduring effects on knowledge, receptivity to knowledge, cognitive flexibility, and cognitive ability (Hyman et al. 1976; Kohn and Schooler 1982; Pascarella and Terenzini 1991). Among adults, years of schooling significantly increase intellectual ability, measured by the Armed Forces Qualification Test, even after adjustments for earlier IQ score (see appendix in Herrnstein and Murray 1994:591, table). Even among children of the same age, small differences in the quantity of schooling significantly increase ability. Quantity of schooling, assessed by the number of hours per day school meets, the number of days in the school year, and the average daily attendance, has large effects on verbal, reading, and math scores (Wiley 1976). The more time children spend in school, the higher their abilities. The strength and significance of the association is striking, given small variations in the quantity of schooling among children of the same age compared with large variations in the quantity of schooling among adults (and given the argument that school quality has little effect [Coleman et al. 1966]).

Schooling builds human capital—skills, abilities, and resources—which ultimately shapes health and well-being. Human capital acquired in school increases a person's real and perceived control over life. Education develops the habits and skills of communication: reading, writing, inquiring, discussing, researching, and reasoning. It develops basic analytic skills such as observing, experimenting, summarizing, synthesizing, interpreting, and classifying. Because education develops one's ability to gather and interpret information and to solve problems on many levels, it increases one's control over events and outcomes in life (Mirowsky and Ross 1989). Moreover, in school, one encounters and solves problems that are progressively more difficult, complex, and subtle. The process of learning builds skills and confidence in problem solving. Education instills the habit of meeting problems with attention, thought, action, and persistence. Thus, education increases effort and ability, the fundamental components of problem solving (Wheaton 1980). Therefore, high levels of education are associated with a sense of personal control (Pearlin et al. 1981; Ross and Mirowsky 1989; Wheaton 1980).

The sense of personal control may be an important link between education and health. Belief in personal control is a learned expectation that outcomes are contingent on one's choices and actions. The individual believes that he or she can master, control, or effectively alter the environment. The sense of personal control appears in a number of related forms, including internal locus of control (Rotter 1966), mastery (Pearlin et al. 1981), instrumentalism (Wheaton 1980), self-efficacy (Gecas 1989), and personal autonomy (Seeman and Seeman 1983). People with a high sense of control report being effective agents in their lives. On the other end of the continuum, people with a sense of power-

lessness believe that their actions do not affect outcomes, but that outcomes are determined by external forces such as powerful others, luck, fate, or chance. According to Seeman (1983), powerlessness is the primary type of subjective alienation.

The sense of personal control improves health partially through health-enhancing behaviors. Compared with people who feel powerless to control their lives, people with a sense of personal control know more about health, are more likely to initiate preventive behaviors like quitting smoking, exercising, or moderating alcohol consumption, and, in consequence, have better self-rated health, fewer illnesses, and lower rates of mortality (Seeman and Lewis 1995; Seeman and Seeman 1983; Seeman, Seeman, and Budros 1988). In contrast, lack of personal control makes efforts seem useless; if outcomes are beyond one's control, why exercise, quit smoking, or avoid overweight?

Education also increases the likelihood of having supportive relationships (Ross and Mirowsky 1989; Ross and Van Willigen 1997). Schooling may promote supportive and equitable relationships because it helps partners understand and negotiate with each other. Education develops cognitive flexibility, which includes the ability to see more than one side of an issue. Inflexible people respond to differences in preferences, opinions, and goals with anger, indignation, and punishment. Flexible people respond with attempts to understand the other's position and to reach mutually satisfactory arrangements, which helps to build and maintain supportive relationships. Social support, in turn, improves health and decreases mortality (House, Landis, and Umberson 1988). Support decreases depression, anxiety, and psychological distress, which erode health (Aneshensel, Frerichs, and Huba 1984; Bruce and Leaf 1989; Kessler and McLeod 1985). Further, supportive partners may encourage exercise or discourage smoking or heavy drinking (Ross, Mirowsky, and Goldstein 1990; Umberson 1987).

Education improves health lifestyle both directly and through perceived control and social support. Compared with those with little schooling, the well educated are more likely to exercise, more likely to drink moderately rather than to abstain or drink heavily, and less likely to smoke; well-educated women are less likely to be overweight (Ross 1994; Ross and Bird 1994; Ross and Wu 1995). In turn, exercise is associated with good health; smoking, being overweight, and drinking heavily or abstaining from drinking are associated with poor health (Berkman and Breslow 1983; Berlin and Colditz 1990; Gaziano et al. 1993; Guralnik and Kaplan 1989; Leon et al. 1987; Manson et al. 1995; National Center for Health Statistics 1989; Paffenbarger et al. 1993; Sandvik et al. 1993; Segovia, Bartlett, and Edwards 1989; Stampfer et al. 1988; Surgeon General 1982; U.S. Preventive Services Task Force 1989).

The quantity model predicts that years of schooling is the crucial factor in the relationship between education and health. Although education improves health by improving work and economic conditions, its benefits are not *solely* due to labor market advantage. Schooling's effects are mediated

partly by the sense of personal control, social support, and health lifestyle.

The Credential Model

According to Collins (1979), education is an inherently worthless credential providing no real skills; but because employers use degrees in hiring decisions, higher education leads to good jobs and high incomes. Employers use degrees to screen applicants in hiring decisions, but education itself is an artificial good, a marker for an ascribed status rather than a real achievement (Collins 1979). Education is largely symbolic because schools do not teach skills necessary for work; most skills are acquired on the job (Collins 1979). Many social scientists speak of education as a mechanism for assigning individuals to socioeconomic positions that legitimizes the resulting inequality (Bowles and Gintis 1976). Some appear to doubt that education adds any value in the process, except to make winners and losers alike see fairness in the outcome (Aronowitz and Giroux 1993; Berg 1970; Bourdieu 1977; Bowles and Gintis 1976).

Much of Collins' view of schooling as an artificial credential is based on a very narrow definition of skill. He equates "skills" with specific job skills, like operating a machine, most of which probably are learned on the job. General cognitive skills learned in school, however, could help a worker read the instructions for machine operation, determine the cause of a machine malfunction, and fix it, or explain to a coworker how the machine works—all of which could increase one's effectiveness on the job. Education's positive effect on performance at work is probably due more to general cognitive skills than to specific job-related skills. Nonetheless, many social scientists accept the claim that education is unrelated to skills. For instance, Burris (1983:455) concludes,

Privileged positions are understood as positions which exercise greater technical skills, and highly educated persons are assumed to be more likely to obtain such positions because they have acquired such skills through education. As Collins and others have demonstrated, this view bears little resemblance to reality.

In contrast to the quantity model, the credential model sees no added benefit to another year of schooling unless it confers a degree. It is the credential that gives a person access to good jobs with high incomes. The credential model makes two predictions. First, the credential of a college degree is the crucial aspect of education. According to Kerckhoff, Campbell, and Trott (1982), support for the credential (or certification) hypothesis would show a significant effect of possessing a college degree, net of the effect of years of education. Second, the degree's benefits are mediated by advantaged labor market positions.

The Selectivity Model

The selectivity model combines aspects of quantity and credential models (Ishida, Spilerman, and Su 1997). Theory and research on selectivity, unlike the quantity and credential

models, is limited. The selectivity of colleges or universities could indicate four factors that are not mutually exclusive: quality of the education, lucrativeness of the college's credential or network, inculcation with the school's culture, and selection for better students.

The first two possibilities view greater selectivity in terms already discussed: greater human capital or more elite credential. First, the selectivity of a college or university could indicate the quality of the education it provides. Students at more selective schools may acquire more extensive and effective information and skills. If so, the additional human capital may produce broad improvements in later work and economic conditions, social psychological resources, and health lifestyle that benefit health. Second, credentials from more selective schools may carry a higher market value, opening doors to more powerful and lucrative positions. The selectivity of a school may indicate the value of the positions to which it gates access. Credentials from selective schools may open the network connections that help one acquire a well-paying, fulfilling job. Granovetter (1974) argues that personal contacts are more important than the human capital acquired with additional years of schooling in helping a person get a high-paying, fulfilling job. Many of these contacts are established in college, and selective schools provide better contacts (Ishida et al. 1997; Rosenbaum 1984). This interpretation suggests that school selectivity affects health because it leads to better jobs with more promotion opportunities and higher incomes.

School culture is a third factor that might produce an association between selectivity and health. Elite and nonelite schools may have different cultures that shape lifestyles and habits of drinking, participating in athletic activities, smoking, and other behaviors that ultimately affect health (Cockerham, Rutten, and Abel 1997). According to Bourdieu (1977), schools perpetuate inequality by rewarding the culture (i.e., behavior, habits, tastes, lifestyles, and attitudes) of the dominant class. This cultural capital could include practices like going to museums, concerts, lectures, and plays; jogging, going to the gym, or participating in sports; and staying slim or avoiding smoking. Strangely enough, Bourdieu claims that cultural capital has no inherent value, but is largely a symbolic marker of being from the right social class. Because the possession of this cultural capital is a sign of being from a higher social class, these children are rewarded by teachers and are encouraged at school; thus they do well in school, complete high school, and go on to college. According to Bourdieu, schools do not teach this culture, but they reward students who learned it at home (DiMaggio 1982). We extend the notion of cultural capital in two ways. First, schools do not simply reward the cultural capital of the higher social classes; they are environments that promote and reinforce habits, lifestyles, and behaviors. The student culture of some schools may encourage drinking, smoking, or athletic participation, whereas others do not. Second, even if jogging or drinking habits act as symbolic markers of class, they may have real effects on health. This interpretation suggests that school selectivity may affect health through lifestyle.

Finally, school selectivity may have no effect independent of the social origins of its students. As implied by the term *selectivity*, prestigious schools may simply select the best students from the most advantaged social backgrounds but provide no added value. If selectivity's effects are spuriously due to the social origins of its students, then school selectivity may have no net association with health after adjustment for indicators of advantaged origins, such as race or parental education.

The selectivity model predicts that school selectivity has a net association with health over and above that predicted from the additional years of schooling or from simply having a college degree. If selectivity has such an association, then adjusting for hypothetical confounders or mediators might suggest the reason. For example, if better work and economic conditions appear to be the sole link between selectivity and health, elite credentials and networks likely produce the association. On the other hand, if health lifestyle appears to be the sole link, cultural capital is likely the key.

MODEL AND ANALYSES

In the primary model we incorporate all three aspects of education (quantity, credential, and selectivity) to determine which is associated with health net of the others. The model predicts health (H_i) from the years of formal schooling completed (Y_i), whether one has a college degree (D_i , scored 0 for less than bachelor's degree and 1 for bachelor's degree and above), and the selectivity of the institution granting the highest degree (S_i , scored from 0 for unselective schools to 7 for the most selective institutions), adjusting for socio-demographic precursors of education (X_i) such as sex, age, and parental education:

$$H_i = b_0 + b_1 Y_i + (b_2 + b_3 S_i) D_i + \sum_{j=4}^k b_j X_{ij} + u_i. \quad (1)$$

Eq. (1) describes a linear relationship of health to years of education in which a college degree may be associated with a discontinuous step up that may be larger the more selective the institution of one's highest degree. The partial derivative of predicted health with respect to years of education is $\partial H_i / \partial Y_i = b_1$. Thus b_1 represents the increment in predicted health attributable to quantity of education apart from that attributable to the credential of a college degree or the selectivity of the college. The partial derivative with respect to having a college degree is $\partial H_i / \partial D_i = b_2 + b_3 S_i$. The function represents the increment in predicted health associated with having a college degree beyond that attributable to the additional years of education alone. In the function, the coefficient b_2 represents the extra increment if the college degree comes from an institution lacking selectivity ($S_i = 0$). In that sense, it represents the purest effect of a college degree unrelated to and unaltered by college selectivity. The partial derivative with respect to the selectivity of the college or university granting the highest degree is $\partial H_i / \partial S_i = b_3 D_i$. The derivative with respect to college selectivity is zero for those who did not go to college ($b_3 \times 0 = 0$; their predicted value does not depend on college selectivity, which is undefined

for them) and b_3 for those who went to college ($b_3 \times 1 = b_3$). Thus b_3 represents the increment in predicted health that is associated with a unit increment in college selectivity for those with a college degree, apart from the increments attributable simply to having a college degree and to the additional years of schooling.

We can estimate the basic model using a product term. Multiplying through Eq. (1) yields Eq. (2), which has a new variable representing the product of college degree by college selectivity ($S_i D_i$). The variable takes the value of zero for persons with no degree, regardless of the filler value assigned to S_i . It takes the value of S_i for persons with a college degree. This does not imply that a degree from an unselective college or university is the same as no college degree, because the term D_i distinguishes between those two statuses (Mirowsky 1999).

$$H_i = b_0 + b_1 Y_i + b_2 D_i + b_3 (S_i D_i) + \sum_{j=4}^k b_j X_{ij} + u_i. \quad (2)$$

A sequence of regression models adjusts for possible confounders or mediators of the relationship between health and the three aspects of education. Model 1 adjusts for race and parental education as well as the individual's age, sex, and marital status. Education's positive association with health may simply reflect advantaged family background (Leigh and Dhiri 1997). Adjustment for race and parental education helps ensure that the estimated effects of respondent's education on health are not spuriously due to advantaged social origins. Subsequent models progressively add adjustment for work and economic conditions (Model 2), social psychological resources (Model 3), and health lifestyle (Model 4). To the extent that the coefficients b_1 , b_2 , and b_3 diminish from Model 1 to Models 2, 3, and 4, the associations they represent result from the work and economic conditions, social psychological resources, and health lifestyle associated with the corresponding aspects of education.

SAMPLE

Our analyses use the 1995 survey of Aging, Status, and the Sense of Control (ASOC), a national telephone probability sample of U.S. households supported by the National Institute on Aging. Sampling, pretesting, and interviewing were conducted by the Survey Research Laboratory of the University of Illinois. Respondents were selected through a prescreened random-digit dialing method that decreases the probability of contacting a business or nonworking number and produces lower standard errors than the standard Mitofsky-Waksberg method while producing a sample with the same demographic profile (Lund and Wright 1994; Waksberg 1978). The survey was limited to English-speaking adults. The ASOC survey has two subsamples designed to produce an 80% oversample of persons aged 60 or older. The main sample drew from all households; the oversample drew only from households with one or more seniors. In the main sample, the adult (aged 18 or older) with the most recent birthday was selected as respondent (O'Rourke and Blair 1983). In the oversample, the senior (aged 60 or older) with the most

recent birthday was selected. Up to 10 callbacks were made to select and contact a respondent, and up to 10 were made to complete the interview once contact was made. Interviews were completed with 71.6% of contacted and eligible persons: 73.0% for the main sample and 67.3% for the over-sample. The final sample has 2,592 respondents ranging in age from 18 to 95. Fifty-eight percent of the sample is under age 60 ($N = 1,496$), and 42% of the sample is 60 years and older ($N = 1,097$). A large number of older persons ensures greater variation in health (Verbrugge 1986). Because the survey oversampled respondents age 60 and older by 1.8, we weighted the sample using a weighting variable equal to 1 if age 59 or younger and 1/1.8 if age 60 or older.

The following statistics compare the demographic characteristics of the ASOC sample to those for the U.S. population as a whole (U.S. Bureau of the Census 1995). The sample statistics are weighted to compensate for the over-sample of seniors. For ASOC and the United States, respectively, the percentage female is 56.2 and 51.2, the percentage white is 85.1 and 82.9, the percentage married (excluding cohabitators and the separated) is 55.7 and 55, and the mean household size is 2.67 and 2.59. For persons age 25 or older the percentage with a high school degree is 85.1 and 80.9, and the percentage with a college degree is 25.6 and 22.2. The mean household income is \$43,949 and \$41,285.

MEASUREMENT

Measuring Education

We measure the quantity, selectivity, and credential of respondents' schooling. Quantity is measured in *years of formal education* completed. *College degree*, a measure of credentials, is a dummy variable coded 1 for respondents who have a college degree or higher, and 0 otherwise. *College selectivity* indicates the status of the institution and, according to Cass and Birnbaum (1991), the academic quality the school. Respondents with a college degree or more were asked the name and location of the institution from which they received their highest degree. Based on this, we scored selectivity of the institution from 0 for unranked schools to 7 for the most selective schools in the United States (Cass and Birnbaum 1991). Selectivity is based on grade point average and SAT/ACT scores of entering freshmen and the percentage of applicants accepted. Examples of the most selective schools include Amherst, Williams, and Swarthmore Colleges, MIT, University of Chicago, Yale, Harvard, Princeton, and Stanford Universities; unranked schools include Central Missouri State University, Southern Arkansas University, Davis and Elkins, Framingham State, Fort Valley State, and North Georgia Colleges. Unranked colleges and universities typically accept the most applicants. They have either low admission criteria (such as a C— high school grade point average) or open admissions. The faculty do not hold doctorates. Selectivity scores are more accurate for persons with a bachelor's degree than for persons with advanced degrees, because the status of an advanced degree depends on the department, not just the school. Although no quality ranking is

unproblematic, inclusion of school selectivity allows us to distinguish between a person who has a degree from Stanford University and one with a degree from the North Georgia College, for example. We use Cass and Birnbaum selectivity scores because the Astin scores are dated (Astin 1971).¹

We measure parents' years of schooling and the respondent's knowledge of parents' schooling. *Parental education* is coded as the number of years of formal schooling completed. It is the average of one's mother's and father's education if both are reported. When one parent's education is unknown, we use the other. *Unknown parental education* is a variable coded 1 if the respondent did not know his/her mother's and father's education; 0.5 if the respondent knew one but not the other; and 0 if the respondent knew both.

The average number of years of formal schooling completed among respondents in our representative sample in the United States is 13.399. Their parents averaged 11.206. About 26% of respondents have a college degree or higher, compared with 10% of their parents. Of the respondents who have a college degree or more, 34.7% attended a private institution (including those with a religious affiliation), and 65.3% went to a public college or university. On a selectivity scale on which 7 is the most selective and 0 is unranked, the average selectivity score is 1.59.

Measuring Health

Health is measured as physical functioning and perceived health. *Physical functioning* assesses physical mobility and functioning in daily activities, using an index of five items similar to Nagi's (1976) disability scale. The questions about physical functioning ask respondents to report relatively specific problems that represent hindrances or limitations. Respondents were asked "How much difficulty do you have (1) climbing stairs; (2) kneeling or stooping; (3) lifting or carrying objects less than 10 pounds, like a bag of groceries; (4) preparing meals, cleaning house or doing other household work; (5) shopping or getting around town?" The response categories are a *great deal of difficulty* (coded 0), *some difficulty* (coded 1), and *no difficulty* (coded 2). Exploratory factor analysis indicates that the five items form a single factor. Averaging the items produces an index with an alpha reliability of .84. The physical functioning score is computed as $100(M^{10}/2^{10})$, where M is the mean of the five items. Raising M to the 10th power corrects the skewness of its distribution, thus minimizing heteroscedasticity in its regressions (Hamilton 1992). Multiplying by $100/2^{10}$ rescales so that scores range from 0 to 100. Low scores indicate physical

1. We also examined a crude indicator of selectivity that distinguishes between private institutions, including those with a religious affiliation (coded 1) and public institutions (coded 0). Typically private colleges and universities in the United States are more selective than are public institutions. (Selectivity and private school correlate at $r = .336$.) When entered simultaneously in the model, however, neither was significant. Because degree of selectivity on an eight-point scale is probably a more refined and precise indicator of quality, we removed the dummy variable for public versus private and retained the measure of selectivity.

impairment or disability; high scores indicate unimpaired physical functioning.

Perceived health is the respondent's subjective assessment of his or her general health as *very poor* (coded 1), *poor* (2), *satisfactory* (3), *good* (4), or *very good* (5). Studies suggest that perceived health is a valid and reliable measure of general physical well-being (Davies and Ware 1981). It combines the subjective experience of acute and chronic, fatal and nonfatal diseases, and general feelings of well-being, like feeling run down and tired and having backaches and headaches. Thus, it measures health as defined by the World Health Organization: as a state of well-being, not simply as the absence of disease. Perceived health correlates highly with more objective measures such as physician's assessments and diagnoses, and it predicts mortality over and above measures of chronic and acute disease, physician's assessment made by clinical exam, physical disability, and health behaviors (Davies and Ware 1981; Idler and Benyamini 1997; Liang 1986). In fact, self-assessed health is a stronger predictor of mortality than is physician-assessed health (Mossey and Shapiro 1982).

Measuring Work and Economic Conditions

We measure *employment status* using four categorical variables: employed full-time, employed part-time, not employed for pay (the comparison group in the regression analyses), and unable to work because of disability or illness. We include inability to work because of disability or illness to control for the selection of some people out of the labor force because of illness; this will allow us to examine whether employment is associated with improved health, holding selection effects constant. *Household income* is coded in thousands of U.S. dollars per year. *Economic hardship* is measured as the response to three questions: "During the past twelve months, how often did it happen that you (1) did not have enough money to buy food, clothes, or other things your household needed; (2) did not have enough money to pay for medical care; and (3) had trouble paying the bills?" Responses to each question were coded from 1 to 4 (1 = *never*, 2 = *not very often*, 3 = *fairly often*, and 4 = *very often*). The economic hardship index is the mean response to the three questions (alpha reliability = .82).

Work fulfillment is measured as fulfillment from work, either paid or unpaid. We asked employed and nonemployed persons to describe the work, tasks, or activities they perform most during the day. Respondents were then asked about the subjective rewards of their primary daily work. Paid work is considered the primary daily work of people working for pay 20 hours per week or more. Unpaid work includes reported activities such as housework, child care, care for an ill or elderly family member, volunteer work, and gardening and home repair. Work fulfillment includes enjoyment of work and the sense of learning and developing as a person through work, measured by responses to two questions: "How much do you agree with the statements: 'My work/tasks give me a chance to do things I enjoy'; and 'My work/tasks give me a chance to develop and to learn new

things?'" (coded 1 = *strongly disagree*, 2 = *disagree*, 3 = *agree*, 4 = *strongly agree*). The index is the average response to the questions.

Measuring Social Psychological Resources

Sense of control is the belief that one can and does master, control, and shape his or her life. Sense of control is measured by a 2 × 2 index that balances statements claiming or denying control over good or bad outcomes (Mirowsky and Ross 1991). The items are (1) "I am responsible for my own successes" and (2) "I can do just about anything I really set my mind to" for *claiming control over good outcomes*; (3) "My misfortunes are the result of mistakes I have made" and (4) "I am responsible for my failures" for *claiming control over bad outcomes*; (5) "The really good things that happen to me are mostly luck" and (6) "There's no sense planning a lot—if something good is going to happen it will" for *denying control over good outcomes*; and (7) "Most of my problems are due to bad breaks" and (8) "I have little control over the bad things that happen to me" for *denying control over bad outcomes*. Responses to control questions (1 through 4) are coded -2 = *strongly disagree*, -1 = *disagree*, 0 = *neutral*, 1 = *agree*, and 2 = *strongly agree*. Responses to lack of control questions (5 through 8) are coded 2 = *strongly disagree*, 1 = *disagree*, 0 = *neutral*, -1 = *agree*, and -2 = *strongly agree*. A mean score sense-of-control index was created from these questions, coded from -2 for a low sense of control to 2 for a high sense of control (alpha reliability = .68). This measure is conceptually similar to the personal control component of Rotter's (1966) locus of control scale (modified for community surveys by using Likert scale responses rather than forced-choice responses) and to Pearlin et al.'s (1981) mastery scale. The major difference is that the scale balances statements claiming control against those denying control and statements about good outcomes against those about bad outcomes (Mirowsky and Ross 1991).

Social support is measured by responses to four questions. The first two assess emotional support, and the second two measure instrumental support. Respondents were asked, "How much do you agree with the statements: 'I have someone I can turn to for support and understanding when things get rough,' 'I have someone I can really talk to,' 'I have someone who would help me out with things, like give me a ride, watch the kids or house, or fix something,' and 'I have someone who would take care of me if I were sick'" (coded 1 = *strongly disagree*, 2 = *disagree*, 3 = *agree*, 4 = *strongly agree*). The social support index is the mean response.

Measuring Health Lifestyle

Walking was measured as the number of days walked per week. Respondents were asked, "How often do you take a walk?" Open-ended responses were coded as the number of days walked per week. *Exercise* is an index of moderate and strenuous exercise. We assessed strenuous exercise by asking respondents, "How often do you do strenuous exercise such as running, basketball, aerobics, tennis, swimming, biking, and so on?" Open-ended responses were coded into

number of days of strenuous exercise per week. We assessed moderate exercise by asking respondents, "How often do you do moderate exercise like golf, bowling, dancing, gardening, or playing games with children?" Open-ended responses were coded into number of days of moderate exercise per week. Our exercise index is the mean response to the questions about moderate and strenuous exercise.

Smoking is coded 0 for persons who never smoked, 1 for persons who have ever smoked seven or more cigarettes a week but who do not currently smoke (quit smoking), and 2 for persons who currently smoke seven or more cigarettes a week (current smoker).

We measured *drinking quantity/frequency* first by asking respondents about frequency of drinking: "On average, how often do you drink any alcoholic beverages such as beer, wine, or liquor?" Frequency was coded in number of days per week based on open-ended responses. Quantity was measured with response to the question, "On the days that you drink, on average, how many alcoholic drinks do you have?" Quantity was coded as the number of drinks per day (from a low of 0 for nondrinkers to a high of 25 drinks per day). We computed quantity/frequency scores by multiplying the number of days per week a person drinks by the number of drinks reported for the average day. Scores were categorized because the effect of drinking on health is likely nonlinear, with both abstainers and heavy drinkers reporting worse health than moderate drinkers. Abstainers reported that they never drink or drink less than one drink a year (44% of the sample). Moderate drinkers drink up to four drinks a day, or 28 drinks per week (54% of the sample). Heavy drinkers drink more than four drinks a day, or 28 drinks per week (2% of the sample). Moderate drinking is the reference category in the regression analysis.²

Degree of overweight was measured by the Body Mass Index (BMI) of weight relative to height (kg/m^2). Of the various weight-relative-to-height measures, weight/height^2 is the most adequate because it is least correlated with height and highly correlated with skin-fold measures indicating body fat (Roche et al. 1981). Degree of overweight, or weight relative to height, is based on self-reported weight and height. Self-reported weights and scale weights are highly correlated; the small error that exists reflects the tendency for people who weigh a lot to underestimate their weights, but the bias introduced by using self-reports is "small and inconsequential" (Palta et al. 1982:230; Stunkard and Albaun 1981).³ We ex-

amine relative weight as a continuous variable, coded from low to high levels of overweight. The average BMI score in this sample is 25.535 overall, 26.239 for men, and 25.040 for women. Thirty percent of the sample scores 27 or higher, which is often used as a cutoff for obesity. The average man is 5'10" and weighs 183.9 pounds; the average woman is 5'4" and weighs 145.9 pounds.

Measuring Sociodemographic Characteristics

Sex is coded 1 for males and 0 for females. *Age* is coded as age in years. *Race* is coded 1 for whites and 0 for nonwhites. *Marital status* is a dummy variable contrasting those currently married or living together as married (coded 1) with those who are single, separated, divorced, or widowed (coded 0).

RESULTS

Basic Model

Of the three aspects of education, quantity (i.e., the years of formal schooling) has the strongest and most significant net association with both physical functioning and perceived health, as shown in Model 1 of Tables 1 and 2. In both tables, the size of the standardized partial regression coefficient associated with years of education is second only to that of age. The credential of a college degree has no significant association with either measure of health, net of years of schooling. College selectivity has a small net association with both physical functioning and perceived health that is statistically significant ($p < .05$) by a one-tailed test, although of borderline statistical significance ($p < .10$) by a two-tailed test. Thus, given the model, the linear association between health and years of education has a modest step up associated with a college degree from a selective institution beyond that predicted from the additional years of education alone, with the size of the step generally increasing with greater selectivity.

Although the measures of the three aspects of education correlate, it is unlikely that multicollinearity accounts for the statistically insignificant coefficients relating health to college degree. College degree has a bivariate correlation of .776 with years of education. (Among the college educated, years of schooling and selectivity correlate at $r = .195$.) Nevertheless, all three measures have tolerance values well above .10. The last column of Table 1 shows the tolerance values for the full set of independent variables, where tolerance is defined as the fraction of variance not predictable from the other independent variables. College degree has a tolerance of .345, which suggests little risk of multicollinearity problems.

Health's association with years of schooling is essentially linear, although perhaps not entirely. It is best approximated as linear in the multivariate model, indicating that each year of schooling is associated with an improvement in

2. We found no differences between rare and moderate drinking. We also used various definitions of heavy drinking. No category of heavy drinking was extreme enough to significantly worsen health in the analyses; any fewer than four drinks a day is associated with good health. Furthermore, only 2% of the sample reported drinking four or more drinks a day, so more extreme cutoffs included very few people. Low levels of drinking reported may be due to the older age of the sample, to a tendency to give socially desirable responses, or to the fact that 98% of respondents in a representative sample really drink less than four drinks a day.

3. Although body weight combines lean body mass and adipose tissue, two facts indicate that most adult variance in weight for height is due to body fat. First, BMI scores increase with age, which indicates that most overweight is due to body fat, given that muscle and bone mass are not

likely to increase as adults age (Ross and Mirowsky 1983). Second, people who exercise strenuously score lower on the BMI than those who do not, even though strenuous exercise increases muscle and bone mass (Hayes and Ross 1986).

TABLE 1. PHYSICAL FUNCTIONING REGRESSED ON RESPONDENT'S EDUCATION, CONTROLLING FOR PARENTS' EDUCATION AND SOCIODEMOGRAPHICS (MODEL 1), WORK AND ECONOMIC CONDITIONS (MODEL 2), SOCIAL PSYCHOLOGICAL RESOURCES (MODEL 3), AND HEALTH LIFESTYLE (MODEL 4): AGING, STATUS AND THE SENSE OF CONTROL SURVEY, 1995

	Model 1		Model 2		Model 3		Model 4		Tolerance
	<i>b</i>	β	<i>b</i>	β	<i>b</i>	β	<i>b</i>	β	
Respondent's Education									
Years of schooling	1.908*** (.477)	.121	.965* (.466)	.061	.781† (.486)	.049	.592 (.458)	.037	.329
College degree	.662 (2.852)	.007	.473 (2.757)	.005	.501 (2.753)	.005	−.606 (2.686)	−.006	.345
College selectivity	1.217† (.702)	.037	1.257† (.678)	.038	1.005† (.676)	.036	.718 (.660)	.022	.700
Work and Economic Conditions									
Employed full-time ^a			7.508*** (1.787)	.090	7.263*** (1.785)	.087	8.958*** (1.752)	.107	.631
Employed part-time ^a			3.104 (2.685)	.022	3.326 (2.678)	.023	3.851 (2.608)	.027	.831
Unable to work ^a			−39.139*** (4.286)	−.165	−37.502*** (4.296)	−.159	−34.838*** (4.198)	−.147	.878
Household income			.002 (.016)	.027	.002 (.016)	.024	.002 (.015)	.026	.885
Economic hardship			−6.892*** (1.178)	−.110	−6.428*** (1.181)	−.103	−5.144*** (1.159)	−.082	.806
Work fulfillment			3.143** (1.255)	.045	1.972 (1.308)	.028	1.642 (1.281)	.024	.820
Social Psychological Resources									
Sense of control					5.791*** (1.563)	.071	5.685*** (1.526)	.070	.781
Social support					1.814 (1.528)	.023	1.622 (1.487)	.020	.804
Health Lifestyle									
Walk							.397* (.207)	.033	.945
Exercise							.664*** (.207)	.057	.877
Smoke							−3.820* (.908)	−.073	.912
Abstain from drinking ^b							−3.138* (1.509)	−.037	.872
Heavy drinking ^b							8.105 (6.103)	.022	.971
Overweight							−1.596*** (.154)	−.179	.927
Parents' Education and Sociodemographics									
Parents' education ^c	.315 (.267)	.024	.215 (.259)	.016	.153 (.259)	.012	−.001 (.253)	−.007	.760
Don't know parents' education ^d	−7.017** (2.260)	−.061	−4.856** (2.183)	−.042	−4.077† (2.185)	−.036	−2.815 (2.131)	−.025	.798
Male	5.758*** (1.523)	.068	3.973** (1.512)	.047	4.070** (1.520)	.048	6.399*** (1.514)	.076	.865
White	1.983 (2.133)	.017	.161 (2.060)	.001	−.336 (2.058)	−.003	−.817 (2.010)	−.007	.920

(continued)

(Table 1, continued)

	Model 1		Model 2		Model 3		Model 4		Tolerance
	<i>b</i>	β	<i>b</i>	β	<i>b</i>	β	<i>b</i>	β	
Age	-.778*** (.048)	-.329	-.761*** (.052)	-.321	-.729*** (.053)	-.308	-.630*** (.053)	-.266	.555
Married	3.350* (1.539)	.040	.742 (1.505)	.009	.121 (1.522)	.001	.933 (1.485)	.011	.900
Constant	67.244		79.850		75.387		114.905		
<i>R</i> ²	.181		.243		.248		.290		

Notes: Numbers in parentheses are standard errors. *N* = 2,493.

^aCompared with not employed (for reasons other than health).

^bCompared with moderate drinking.

^cScored in number of years of formal education completed. Average of mother's and father's education.

^dScored as *know both parents* = 0, *don't know one* = 0.5, *don't know either* = 1.

[†]*p* < .10; **p* < .05; ***p* < .01; ****p* < .001 (two-tailed tests)

health. In a supplementary analysis (not shown), we replaced years of schooling with dummy variables representing degrees (less than high school diploma, high school diploma, some college, bachelor's degree or more). We found no significant deviation from linearity in the mean levels of physical functioning and perceived health across these categories, and the explained variance was consistently lower than when we used only years of schooling as the measure of education. Nonetheless, the crude means in Figure 1 suggest a leveling off of physical functioning and perceived health at the master's degree level. Only about 8% of people in our representative sample, however, hold advanced degrees.

The quantity of schooling completed appears to be the most important aspect of education for health. The standardized partial regression coefficients are several times larger

for years of education than for college selectivity. With physical functioning dependent, the standardized coefficients for years and selectivity are .121 and .037, respectively. With perceived health dependent, they are .150 and .042, respectively.

Possible Mediators

High levels of education are associated with full-time employment, fulfilling work, high household income, and high levels of personal control and social support, as shown in Table 3. In contrast, persons with little schooling experience greater economic hardship, and have lower average personal control and social support, and are less likely to be employed. Table 3 shows that compared with those with less schooling, the well educated are more likely to walk, to ex-

TABLE 2. PERCEIVED HEALTH REGRESSED ON RESPONDENT'S EDUCATION CONTROLLING FOR PARENTS' EDUCATION AND SOCIODEMOGRAPHICS (MODEL 1), WORK AND ECONOMIC CONDITIONS (MODEL 2), SOCIAL PSYCHOLOGICAL RESOURCES (MODEL 3), AND HEALTH LIFESTYLE (MODEL 4): AGING, STATUS AND THE SENSE OF CONTROL SURVEY, 1995

	Model 1		Model 2		Model 3		Model 4	
	<i>b</i>	β	<i>b</i>	β	<i>b</i>	β	<i>b</i>	β
Respondent's Education								
Years of schooling	.050*** (.011)	.150	.027** (.010)	.080	.025** (.010)	.073	.020* (.010)	.058
College degree	.020 (.067)	.009	.013 (.064)	.006	.008 (.063)	.004	-.002 (.062)	-.001
College selectivity	.029 [†] (.016)	.042	.030* (.015)	.042	.029 [†] (.015)	.041	.018 (.015)	.025
Work and Economic Conditions								
Employed full-time ^a			.087* (.042)	.047	.086* (.042)	.046	.127** (.042)	.067

(continued)

(Table 2, continued)

	Model 1		Model 2		Model 3		Model 4	
	<i>b</i>	β	<i>b</i>	β	<i>b</i>	β	<i>b</i>	β
Employed part-time ^a			.031 (.062)	.010	.037 (.062)	.011	.049 (.061)	.015
Unable to work ^a			-.968*** (.097)	-.190	-.941*** (.097)	-.184	-.870*** (.095)	-.170
Household income			.001 (.000)	.025	.000 (.000)	.022	.000 (.000)	.019
Economic hardship			-.135*** (.028)	-.095	-.122*** (.028)	-.086	-.103*** (.028)	-.073
Work fulfillment			.214*** (.029)	.138	.178*** (.030)	.114	.167*** (.030)	.107
Social Psychological Resources								
Sense of control					.101** (.036)	.056	.095** (.036)	.053
Social support					.102** (.036)	.056	.094** (.035)	.052
Health Lifestyle								
Walk							.015** (.005)	.057
Exercise							.023*** (.005)	.088
Smoke							-.084*** (.021)	-.072
Abstain from drinking ^b							-.137*** (.035)	-.074
Heavy drinking ^b							-.108 (.136)	-.014
Overweight							-.022*** (.004)	-.112
Parents' Education and Sociodemographics								
Parents' education ^c	.014* (.006)	.049	.010† (.006)	.037	.009† (.006)	.030	.003 (.006)	.012
Don't know parent's education ^d	-.204*** (.051)	-.085	-.161*** (.049)	-.067	-.145** (.049)	-.060	-.109* (.048)	-.045
Male	-.016 (.036)	-.008	-.036 (.035)	-.019	-.028 (.035)	-.015	-.003 (.035)	-.002
White	.162** (.052)	.061	.117* (.049)	.044	.106* (.049)	.040	.105* (.048)	.039
Age	-.009*** (.001)	-.173	-.009*** (.001)	-.183	-.008*** (.001)	-.162	-.006*** (.001)	-.114
Married	.092** (.036)	.049	.034 (.035)	.018	.010 (.035)	.006	.019 (.035)	.010
Constant	3.578		3.506		3.217		3.792	
R ²	.118		.204		.209		.244	

Notes: Numbers in parentheses are standard errors. $N = 2,493$.

^aCompared with not employed (for reasons other than health).

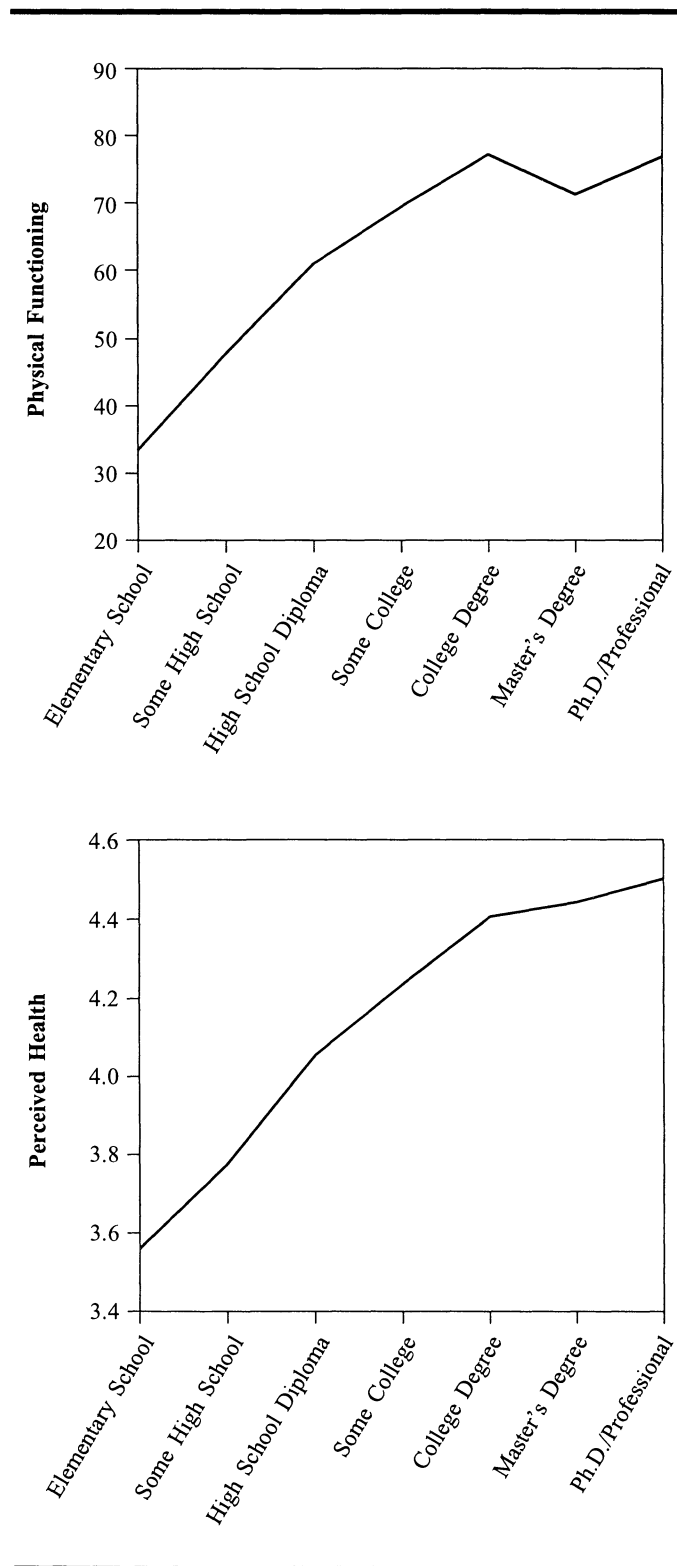
^bCompared with moderate drinking.

^cScored in number of years of formal education completed. Average of mother's and father's education.

^dScored as *know both parents* = 0, *don't know one* = 0.5, *don't know either* = 1.

† $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$ (two-tailed tests)

FIGURE 1. MEAN LEVELS OF PHYSICAL FUNCTIONING AND PERCEIVED HEALTH, BY LEVEL OF EDUCATIONAL ATTAINMENT



ercise, and to drink moderately, and are less likely to abstain from drinking, to smoke, and, among women, to be overweight. (Heavy drinking is not significantly associated with educational attainment). Thus part of the association between education and health may result from education's links to work and economic conditions, social psychological resources, and health lifestyle.

Adjustments for work and economic conditions are added in Model 2 of Tables 1 and 2. People who are employed full-time report significantly better physical functioning and perceived health than those who are not employed, although part-time employment is not significant. Not surprisingly, people who are not working because of disability report much worse physical functioning and perceived health than others. Economic hardship is associated with poorer physical functioning and perceived health. With adjustment for economic hardship, as shown in the tables, household income's coefficient is not significant. (Without adjustment for economic hardship, income is significantly associated physical functioning and perceived health: $p = .006$ and $.035$, respectively.) Work fulfillment has a significant and positive net association with both measures of health. Adding the adjustment for work and economic conditions reduces the coefficient of years of education markedly but does not eliminate it. With physical functioning dependent, the adjustment reduces the coefficient for years of education by about 49%. With perceived health dependent, the adjustment reduces the coefficient by about 46%. The adjustment does not reduce the coefficient associated with college selectivity in either table.

Model 3 adds adjustment for social psychological resources. People with a high sense of control over their lives have significantly better health by both measures. People with high levels of social support have significantly greater perceived health but not significantly better physical functioning. The added adjustment somewhat reduces the coefficient associated with years of education when predicting physical functioning but only slightly when predicting perceived health.

Model 4 adds adjustment for health lifestyle in Tables 1 and 2. People who walk, exercise, refrain from smoking, are not overweight, and drink moderately rather than abstain have better physical functioning and perceived health. (Abstaining from alcohol, compared with drinking moderately, is associated with significantly worse health by both measures.) Adding the adjustment for health lifestyle further reduces the coefficient associated with years of schooling in both tables. Years of schooling becomes statistically insignificant in predicting physical functioning but remains significant in predicting perceived health. Together with work and economic conditions and social psychological resources, lifestyle explains much of the association between respondent's educational attainment and health.

Adding the adjustment for health lifestyle makes the coefficient of college selectivity statistically insignificant in both tables. Thus the association of college selectivity with health lifestyle apparently accounts for much of its association with physical functioning and perceived health. Earlier

TABLE 3. MEAN WORK AND ECONOMIC CONDITIONS, SOCIAL PSYCHOLOGICAL RESOURCES, AND HEALTH LIFESTYLE AT THREE LEVELS OF EDUCATION: AGING, STATUS AND THE SENSE OF CONTROL SURVEY, 1995

	Less Than High School Diploma	High School Diploma to Some College	College Degree or More
Work and Economic Conditions			
Employed full-time ^a	.219 (.414)	.413 (.493)	.549 (.498)
Employed part-time ^a	.050 (.217)	.110 (.313)	.069 (.254)
Not employed ^a	.731 (.440)	.477 (.500)	.382 (.486)
Unable to work ^a	.084 (.277)	.033 (.179)	.008 (.087)
Household income (in \$1,000s per year) ^a	25.671 (25.818)	39.446 (46.845)	59.062 (52.458)
Economic hardship ^a	1.588 (.845)	1.420 (.646)	1.233 (.489)
Work fulfillment ^a	3.015 (.574)	3.114 (.588)	3.330 (.583)
Social Psychological Resources			
Sense of control ^a	.354 (.470)	.625 (.510)	.845 (.466)
Social support ^a	3.142 (.454)	3.276 (.510)	3.399 (.524)
Health Lifestyle			
Walk ^b	3.225 (3.550)	3.507 (3.489)	3.742 (3.357)
Exercise ^a	2.300 (3.442)	3.265 (3.530)	3.565 (3.528)
Smoke ^a	.807 (.802)	.756 (.805)	.517 (.701)
Abstain from drinking ^a	.676 (.469)	.447 (.497)	.312 (.464)
Moderate drinking ^a	.311 (.463)	.537 (.499)	.665 (.472)
Heavy drinking	.008 (.088)	.014 (.118)	.020 (.139)
Overweight ^c	26.330 (5.175)	25.589 (4.552)	24.950 (4.429)

Notes: Numbers in parentheses are standard deviations. $N = 2,493$.

^aEducation categories are significantly different at $p < .001$ (two-tailed tests).

^bEducation categories are significantly different at $p < .10$ (two-tailed tests).

^cThe significant ($p < .001$) association of education with overweight holds for women only, with the means at the three educational levels, from low to high respectively being 26.462, 25.109, and 23.865. For men, there is no significant association of education with overweight. The means at the three educational levels are 26.088, 26.294, and 26.194.

we developed various hypotheses concerning the ways in which the selectivity of higher education might affect health, including labor market advantages and health lifestyle. The results in Table 2 suggest that lifestyle, not work and economic conditions, mediates the effect. Among the college educated, the higher the selectivity of the school one attended, the higher one's household income ($r = .126$) and the less one's economic hardship ($r = -.149$). Thus there appears to be an economic advantage to attending a high-status school. This economic advantage, however, does *not* explain the association between college status and perceived health. Adjusting for work and economic conditions in Model 2 leaves the association between college selectivity and health intact. (Social psychological resources do not explain the association between school selectivity and health, as adjustment for them in Model 3 leaves the association largely unchanged). Only when health lifestyle is adjusted does the association between college selectivity and perceived health diminish substantially. Among the college educated, those who attended more selective schools are significantly more likely to walk and exercise and are less likely to be overweight or to abstain from drinking ($r = .077, .081, -.080$, and $-.135$, respectively). Multiple regression analyses (not shown) indicate that these relationships are significant when we adjust for parental education and sociodemographic characteristics (age, sex, race, and marital status). Graduates from high-status schools, however, are no less likely to smoke than those from lower-status schools ($r = .019$). All college graduates, no matter what the status of their institution, are less likely to smoke than people with less education. Thus, the selectivity of one's degree-granting institution is associated with perceived health and physical functioning because it is associated with a lifestyle that includes walking, exercising, avoiding overweight, and drinking moderately (rather than abstaining), all of which are associated with good health.

College selectivity does not appear to act solely as a stand-in for advantaged background. The coefficient for selectivity in Model 1 is significant when we adjust for race and parental education. Removing race and parental education from the equation increases the selectivity coefficient somewhat, consistent with the argument that selective schools tend to accept applicants from advantaged backgrounds. When predicting physical functioning, the coefficient is 1.335 ($t = 1.907$) without those variables adjusted, compared with 1.217 ($t = 1.733$) in Model 1 of Table 1. The coefficient for selectivity is 0.032 ($t = 2.069$) when predicting perceived health without adjustments for race and parental education, compared with 0.029 ($t = 1.774$) in Model 1 of Table 2. Thus advantaged background appears to contribute to a positive correlation between college selectivity and health, but not as much as health lifestyle.

Subsidiary Findings

Age, sex, race, and marital status are controlled in all analyses to ensure that education's positive effect on health is not spuriously due to younger age, male gender, white race, or marriage. Older persons reported poorer physical function-

ing and perceived health than younger persons. Age is the only variable included here that has a larger association with health than does years of education (see Model 1 in Tables 1 and 2, before adjustment for explanatory variables). Married persons report better physical functioning and perceived health than the nonmarried, almost entirely due to better work and economic conditions. Compared with women, men report significantly better physical functioning, but not better perceived health. The difference in physical functioning decreases with adjustment for work and economic conditions, but increases with adjustment for health lifestyle. Whites report significantly better perceived health than nonwhites, but not better physical functioning. Adjustment for work and economic conditions accounts for a large fraction of the difference in perceived health, but not most of it.

We adjust for parental education in all models to ensure that education's effect is not spuriously due to advantaged social origins. Parental education is significantly associated with better perceived health but not better physical functioning.⁴ (Not knowing parental education is significantly associated with poorer health by both measures.) Furthermore, respondent's education has much larger effects on health than does parents' education. Comparison of standardized coefficients in Model 1 of Tables 1 and 2 shows that the effect of one's own educational attainment on both aspects of health is at least three times as large as the effect of one's parents' education ($\beta = .121$ compared with $.024$ for physical functioning, and $\beta = .150$ compared with $.049$ for perceived health). The addition of lifestyle characteristics reduces the effect of parental education on perceived health to insignificance. One of the ways parental education affects adult health is through lifestyle. Children may learn habits of nutrition, exercise, smoking, and drinking at home. These lifestyles, passed on from parents to children, affect adult health. Adding the adjustment for health lifestyle eliminates the association between parents' education and perceived health. Multiple regression analyses (not shown) indicate that even after adjustments for respondent's education and sex, race, and age, the higher one's parents' education, the less likely one is to smoke, to be overweight, and to abstain from drinking rather than drink moderately, and the more likely one is to walk. Well-educated parents appear to transmit a healthful lifestyle to their adult children.

DISCUSSION

Years of formal education are significantly associated with perceived health and physical functioning over and above family background measured as parental education. Years of formal education have a standardized association with perceived health and physical functioning second in size only to that of age. The credential of a degree beyond years of educational attainment has no significant effect. The selectivity of the school attended, however, has a small positive

association with physical functioning and perceived health. The results are consistent with the argument that years of schooling develop effective capacities that shape many of life's circumstances that ultimately affect health, including paid full-time employment, fulfilling work, economic well-being, a sense of personal control and social support, and a healthful lifestyle.

In the 1960s Coleman and others argued that in the United States family background alone, not the quality of schools, shapes children's outcomes (Coleman et al. 1966). His conclusions were based on problematic data analysis in which he decomposed variance rather than examined associations. Nevertheless, the argument continues to be advanced that equal access to high-quality schooling will not overcome family background (Herrnstein and Murray 1994). Our results indicate that quality may have an effect on health to the extent that more selective colleges and universities promote more healthful lifestyles, but that the selectivity of the college appears much less important to health than the additional years of education. Coleman might have been more optimistic about the benefits of schooling if he had considered quantity rather than quality of schooling. The number of years of formal education that a person completes shapes many aspects of life that affect physical well-being, even after adjustments for family background. The more years of schooling a person has, the better his or her health.

The beneficial effects of the years of college schooling on health are even more striking when one considers the modest average quality of the schools. The average selectivity score of the degree-granting colleges and universities is 1.6 on a scale ranging from 0 to 7. More than half of the respondents with a bachelor's degree or higher received the degree from an unselective school. Although the quality of the education at a school with a score of less than 2 may be fairly low, it appears better for well-being to have this education than not to have it. A person who attends an unselective public school probably is better off than a person of similar background who does not attend college. Perhaps unselective schools improve critical skills such as reading and mathematics, even if they fall short compared with selective schools on advanced skills in writing, calculus, logic, and so on. Some people deride the quality of unselective public education in the United States (Henry 1994), but even a low-quality college education may improve knowledge, skills, abilities, and habits. This may account for the apparent benefit to health of the additional years of schooling even at unselective schools.

The credential of the college degree is less important to health than are years of schooling. In fact, over and above the years of formal education completed, a degree confers no health benefit. According to some, college education is largely symbolic—a credential with no inherent value (Collins 1979). In this view, 15 years of school is no better than 14 or 13 without the credential. This perspective appears to have little relevance to physical well-being: Each year spent in school up to 16 is associated with better perceived health and physical functioning. The leveling off of health benefits for people who continue past a bachelor's de-

4. We could not reliably distinguish the effects of mother's and father's education. When the two were entered separately, father's education was somewhat more significant for both men and women; when entered together, neither was significant.

gree but do not attain a Ph.D. or professional degree requires further examination.

In terms of health, we find no evidence of drawbacks to increasing years of schooling through the attainment of a college degree. In contradiction to critics like Berg (1970), who claims that college education does not give one access to fulfilling work, the data imply clear work advantages to finishing high school, attending college, and finishing college. People who decry the unemployment rate among the college educated or claim that higher education does not give people access to fulfilling, well-paying work should ask, compared with whom? We find that the more years of schooling a person has, the more likely he or she is to be employed full-time at a fulfilling job that pays enough so that felt economic hardships are low, apparently improving physical function and perceived health. Moreover, education seems to boost people's sense of controlling their lives and having supportive relationships, perhaps by improving problem-solving capacities and cognitive flexibility. Education is further associated with a lifestyle that includes walking, exercising, drinking moderately, avoiding overweight, and not smoking, all of which significantly improve health. If there are drawbacks to higher education, we do not see any in the realm of physical well-being or the social determinants of physical well-being.⁵ Education's associations with health are uniformly positive.

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5. Education benefits health in all age groups, but its effect is larger among older adults. In part, this pattern is due to aging (Ross and Wu 1996). However, the pattern could also be the result of a cohort effect. We cannot separate aging and cohort effects here because of the cross-sectional nature of our data, but understanding what aspect of education has the greatest effect in different age groups and whether differential effects represent cohort differences or aging could be a fruitful area for research.

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